



Oticon hearing aid

but special circuitry in the speech processor reduces unwanted background noise.

Hearing aids. The hearing aid is a miniature public address system worn by the listener. It works best in quiet, structured settings, where the speaker is no more than a few metres away and extraneous noise is minimised.

Hearing aids are generally available in four styles: body-worn, behind-the-ear, eyeglass and in-the-ear. School-age children most often use postauricular hearing aids, which are designed to fit unobtrusively behind the ear. Almost all people with hearing loss, including 'nerve loss,' can benefit to some extent from hearing aids.

More about hearing aids

Hearing aid technology is constantly getting better. Many hearing aids today have special microphones and amplifiers. Many have small digital circuits inside. This technology affects how the hearing aid amplifies sounds. Many hearing aid manufacturers use their own trade names to label circuits, but some generic terms that apply to hearing instrument circuits are analogue, digital and programmable.

Analogue, digital and programmable. Analogue hearing aids amplify the sound wave by simply making it louder. These use simple components like transistors in a circuit to amplify and modify the incoming sound. Any change to the sound of the hearing aid is made with the volume control,

small trimmers to adjust the response, or by sending the instrument back to the manufacturer to have components changed.

Due to constant improvements in microcontrollers, most of the new hearing aids have digital circuits. This lets the audiologist make computerised adjustments to many different features. The audiologist uses a computer connection to the hearing aid to adjust the hearing aid response, listening programmes and type of processing in the hearing aid.

A true digital hearing aid, based on a digital signal processor, takes the incoming signal from the microphone, converts it into a digital format and then processes the signal using digital technology before converting it back into an analogue sound to be delivered to your ear. Digital hearing aids are normally much more flexible and can be used to process sound more selectively than analogue ones.

Programmable (analogue/digital) hearing aids can have their response (sound) changed in the dispenser's office, so one need not send it back to the factory. If the user needs more power—a change in frequency—the hearing aid fitter can plug a cord into his hearing aid and modify it on the field.

Analogue types are sometimes more powerful than digital gadgets. Besides, their batteries usually last longer and these cost much less than digital. Some digital hearing aids can have multiple memories for different listening situations and certain noise-reduction algorithms can be programmed into the circuit to help reduce background noise. Built-in digital feedback reduction circuits make more gain available without any whistling. Programmable types are less expensive than digital instruments. The listener can adjust the sound to better fit his own likes and dislikes.

Hearing and amplification

Dynamic range compression. When a hearing aid uses compression, the

circuit amplifies softer sounds more than louder sounds. If a hearing aid circuit has a wide dynamic range of compression, it automatically adjusts the amount of gain so that soft sounds are made louder and loud sounds are not distorted or too loud. This kind of circuit may help one hear conversations at different listening distances.

Directional microphones. Some hearing aids have microphone settings that change between omni-directional and directional. Omni-directional microphones pick up sounds coming from all around. Directional microphones pick up sounds from a narrow listening direction, just like a camera that has a wide-angle and zoom lens.

Multiple memories. Many 'true digital' hearing instruments can keep more than one listening program stored in the hearing aid circuit. Multiple programs let the hearing aid user choose different microphone or hearing aid settings by using a remote control or program button on the hearing aid. Adults often use different programs for listening in noisy situations or when listening to TV or music. Having the choice of different programs can be good for children, especially if their hearing fluctuates or changes. This can allow a fast change in the hearing aid settings.

Availability

There are a variety of electronic devices in the market to support or enhance the hearing capability. These could range from very low-technology to very high-technology, with price ranging from a few hundred rupees to several thousand rupees. Some devices are dedicated, meaning that these only provide a means of communication. Others are designed to work in conjunction with computer systems that play multiple roles. ●

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